

Anthony Essel Prepeh

Engineering Project Portfolio

Geological Engineering

Mining Technology

Artificial Intelligence

ENGINEERING PROJECT PORTFOLIO

This portfolio presents selected engineering, research and software development projects that demonstrate my technical knowledge, analytical ability, problem-solving skills and commitment to continuous professional development.

PROJECT 1

Career Operating System

Project Category

Software Development • Career Automation

Project Status

In Progress

Overview

Designed and developed a Python-powered Career Operating System that automatically generates a professional portfolio website, ATS résumé, curriculum vitae, cover letters and supporting career documents from a single source of truth using structured YAML data and reusable templates.

Purpose

To eliminate repetitive document editing, maintain consistency across professional documents and demonstrate the practical application of software engineering principles to career management.

Key Features

- Automated website generation
- Automated résumé generation
- Automated CV generation
- Modular HTML templates

- YAML-driven content management
- Reusable Python rendering system
- Git version control
- Responsive portfolio website

Technologies

- Python
- HTML5
- CSS3
- Markdown
- YAML
- Git
- GitHub
- Pandoc
- Visual Studio Code

Skills Demonstrated

- Software Development
- Workflow Automation
- Technical Documentation
- Version Control
- Web Development
- Problem Solving
- System Design

Repository

<https://github.com/SaintNova327/Anthony-Essel-Prepeh-Career-Package>

PROJECT 2

Orebody Modelling & Resource Evaluation

Project Category

Geological Engineering Research

Project Status

In Progress

Overview

Final-year Geological Engineering project involving geological interpretation, orebody modelling and resource evaluation to support mineral exploration and mine planning.

Objectives

- Interpret geological datasets.
- Construct geological models.
- Support mineral resource evaluation.
- Produce professional technical documentation.

Software

- Leapfrog Geo
- GIS
- Microsoft Excel

Skills Demonstrated

- Geological Interpretation
 - Orebody Modelling
 - Resource Evaluation
 - Engineering Analysis
 - Technical Reporting
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PROJECT 3

Groundwater Analysis of Heavy Metals

Project Category

Hydrogeology

Project Status

In Progress

Overview

Hydrogeological investigation assessing groundwater quality through laboratory analysis of heavy metals and interpretation of environmental data.

Objectives

- Assess groundwater quality.
- Interpret laboratory data.
- Evaluate environmental implications.
- Produce technical reports.

Skills Demonstrated

- Hydrogeology
- Environmental Assessment
- Water Quality Analysis

- Data Interpretation
 - Technical Reporting
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CONTINUOUS PROFESSIONAL DEVELOPMENT

Alongside my academic training, I continuously develop skills in software engineering, automation, artificial intelligence and digital technologies that complement modern Geological Engineering practice.

Current areas of development include:

- ArcGIS Pro
- Leapfrog Geo
- QGIS
- Python Automation
- Artificial Intelligence for Mining
- Engineering Data Analysis
- Technical Documentation